

stellar labels θ

fixed grid τ_{std}

same labels, depth grid, solver, and convergence criterion

(a) six-field neural initializer

six-field
network

predicts $m, T, P_{\text{gas}}, n_{\text{e}}, \kappa_{\text{R}}, g_{\text{rad}}$ independently

(b) learned two-field initializer

two-field
network

predicts m, T

physical synchronization at fixed (m, T)
EOS, opacity, transfer, hydrostatic closure

(c) hydrostatic grey-convective initializer

grey atmosphere
hydrostatic balance
convective adjustment

$m_{\text{seed}} = m_{\text{grey}}$
 $T_{\text{seed}} = T_{\text{conv}}$
 $P_{\text{seed}} = g m_{\text{grey}}$

$\kappa_{\text{seed}} =$
 $\kappa(T_{\text{conv}}, P_{\text{seed}})$
no atmosphere emulator

same atmosphere solver

radiative transfer
equation of state
hydrostatic closure
energy correction

stop when
 $\max_{\text{deep}} |\Delta T/T|$
 $< 5 \times 10^{-4}$

The convective temperature correction does not trigger a second integration of the column-mass coordinate.